FARM MANAGEMENT SYSTEM

Project Report

1. INTRODUCTION

1.1 Project Overview

The Farm Management System is a Salesforce-based application designed to manage important farm-related information in a centralized digital platform. It helps manage farmers, farms, crops, and buyers while reducing manual work and improving the accuracy of records.

1.2 Purpose

The main purpose of the system is to digitize farm data, improve data accuracy, automate record management, and provide real-time monitoring of important farm information.

2. IDEATION PHASE

2.1 Problem Statement

PS-1 – Farmer: Farmers need a simple way to maintain and access their farm and crop information.

PS-2 – Farm Manager: Farm managers need centralized records to monitor farms, crops, and farmers efficiently.

PS-3 – Buyer: Buyers need reliable information about available crops and farmer details.

PS-4 – Admin: Administrators need secure control over users, records, reports, and system access.

2.2 Empathy Map

The empathy map identifies the users' needs, challenges, expectations, and experiences. It helps understand what farmers, farm managers, buyers, and administrators think, feel, say, and do while using the system.

2.3 Brainstorming

Brainstorming was used to identify possible solutions such as centralized record management, automated data entry, role-based access, field validation, reports, dashboards, and real-time monitoring.

3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

The customer journey describes the stages followed by users from entering the system to creating, updating, monitoring, and reporting farm information. It helps identify user needs and areas where the system can improve the experience.

3.2 Solution Requirements

Functional Requirements

FR-1: Create and manage farm records.

FR-2: Create and manage crop records.

FR-3: Create and manage farmer records.

FR-4: Create and manage buyer records.

FR-5: Maintain relationships between related records.

FR-6: Control access using profiles, roles, and sharing rules.

FR-7: Generate reports and dashboards for monitoring.

Non-Functional Requirements

The system should provide security, reliability, usability, performance, scalability, and data accuracy. It should also be easy for authorized users to access and maintain.

3.3 Data Flow Diagram

The Data Flow Diagram represents how information moves between users and the Farm Management System. Farmers and farm managers provide farm, crop, and farmer information. Buyers access relevant buyer and crop information, while administrators manage users, access, reports, and dashboards.

3.4 Technology Stack

Platform: Salesforce Developer Org

Application Platform: Salesforce Lightning Platform

User Interface: Salesforce Lightning App and Tabs

Data Management: Salesforce Objects, Fields, Relationships, Validation, and Reports

Security: Profiles, Roles, and Sharing Rules

4. PROJECT DESIGN PHASE

4.1 Problem-Solution Fit

The proposed system addresses the major problems of manual record keeping by providing centralized data storage, controlled access, automated record management, and reporting capabilities.

4.2 Proposed Solution

The proposed solution is a Salesforce Farm Management System containing four main objects: Farm, Crop, Farmer, and Buyer. The system uses relationships, field dependencies, user access controls, reports, and dashboards to manage farm information.

4.3 Solution Architecture

The architecture consists of users interacting with the Salesforce Lightning application. The application uses Salesforce objects and fields for data storage, relationships for connected information, security controls for access management, and reports and dashboards for analysis and monitoring.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Product Backlog: A prioritized list of project requirements and development tasks.

Sprint Schedule: Development work is divided into manageable sprints with defined goals.

Estimation: Tasks are estimated according to complexity and expected effort.

Project Tracker: Used to monitor task progress and completion.

Velocity & Burndown Chart: Used to track sprint progress, completed work, and remaining work.

6. PROJECT DEVELOPMENT – MILESTONES

Milestone 1 – Salesforce Developer Org Setup

Create and configure the Salesforce Developer Org required for developing and testing the Farm Management System.

Milestone 2 – Objects

Create four main objects: Farm, Crop, Farmer, and Buyer.

Milestone 3 – Tabs Creation

Create tabs for Farm, Crop, Farmer, and Buyer objects for easy navigation.

Milestone 4 – Lightning App

Create the **Farm Management System** Lightning App and configure the required navigation items.

Milestone 5 – Fields and Relationship

Activity 1 – Farm Object Fields: Create fields required to store farm information.

Activity 2 – Crop Object Fields: Create fields required to store crop information.

Activity 3 – Farmer Object Fields: Create farmer information fields and configure the field dependency **State** → **District**.

Activity 4 – Buyer Object Fields: Create fields required to store buyer information.

Milestone 6 – Profile

A profile controls the permissions and access available to a Salesforce user. Create a **Farm Manager Profile** with the required permissions.

Milestone 7 – Role

Create a role to establish the required organizational hierarchy and record visibility.

Milestone 8 – User

Create the **Farm Manager User** and **Sales Department User** with appropriate profiles and roles.

Milestone 9 – Sharing Rules

Configure record sharing using owner-based and criteria-based sharing rules. Create the required sharing rule to provide access to appropriate users.

Milestone 10 – User Adoption

Test the system by creating records, such as a Farmer record, and verifying that users can perform their assigned tasks.

Milestone 11 – Reports

A report is a structured view of Salesforce data used for analysis and monitoring. Create a report for the Farm Management System.

Milestone 12 – Dashboard

A dashboard provides visual summaries of report data. Create a dashboard to display important farm management information.

Milestone 13 – View Report and Dashboard

Verify that authorized users can view and use the generated reports and dashboards.

7. FUNCTIONAL & PERFORMANCE TESTING

Testing Activities

Creation of Records: Verify that Farm, Crop, Farmer, and Buyer records can be created successfully.

Validation of Fields: Check required fields, valid values, and field validation rules.

Relationship and Dependency Checks: Verify object relationships and the State → District field dependency.

Profile and Role Access Testing: Confirm that users receive only the permissions assigned to their profiles and roles.

Sharing Rule Verification: Verify that records are shared with the correct users according to configured rules.

Report and Dashboard Generation: Verify that reports and dashboards display correct and useful information.

8. ADVANTAGES & DISADVANTAGES

Advantages

Centralized System: Farm information is maintained in one platform.

Reduced Paperwork: Digital records reduce dependence on manual documents.

Improved Accuracy: Validation and structured fields help reduce data-entry errors.

Disadvantages

Requires Internet: The system generally requires an internet connection for access.

Salesforce License: Access to Salesforce features may depend on the available Salesforce edition and licenses.

Limited Offline Access: Offline functionality may be limited compared with a dedicated offline application.